

| Tool Version : Vivado v.2025.2 (win64) Build 6299465 Fri Nov 14
19:35:11 GMT 2025
| Date : Sat Jul 18 22:57:23 2026
| Host : AK_LAPTOP running 64-bit major release (build
9200)
| Command : report_utilization -file
mat_accel_utilization_synth.rpt -pb mat_accel_utilization_synth.pb
| Design : mat_accel
| Device : xc7a12tcs325-1
| Speed File : -1
Design State : Synthesized

Utilization Design Information

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1. Slice Logic
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Site Type	Used	Fixed	Prohibited	Available
Util%				
Slice LUTs*	439	0	0	8000
5.49				
LUT as Logic	439	0	0	8000
5.49				
LUT as Memory	0	0	0	5000
0.00				
Slice Registers	74	0	0	16000
0.46				
Register as Flip Flop	74	0	0	16000
0.46				
Register as Latch	0	0	0	16000
0.00				
F7 Muxes	0	0	0	7300
0.00				
F8 Muxes	0	0	0	3650
0.00				
Unique Control Sets	2		0	3650
0.05				

* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.

Warning! LUT value is adjusted to account for LUT combining.

Warning! For any ECO changes, please run place_design if there are unplaced instances

** Note: Available Control Sets calculated as Slice * 1, Review the Control Sets Report for more information regarding control sets.

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	—	—	—
0	—	—	Set
0	—	—	Reset
0	—	Set	—
0	—	Reset	—
0	Yes	—	—
0	Yes	—	Set
0	Yes	—	Reset
1	Yes	Set	—
73	Yes	Reset	—

2. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	0	0	0	20	0.00
RAMB36/FIFO*	0	0	0	20	0.00
RAMB18	0	0	0	40	0.00

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

3. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	0	0	0	40	0.00

4. IO and GT Specific

Site Type	Used	Fixed	Prohibited	Available	Util%
Bonded IOB	136	0	0	150	90.67
Bonded IPADs	0	0	0	10	0.00
Bonded OPADs	0	0	0	4	0.00
PHY_CONTROL	0	0	0	3	0.00
PHASER_REF	0	0	0	3	0.00
OUT_FIFO	0	0	0	12	0.00
IN_FIFO	0	0	0	12	0.00
IDELAYCTRL	0	0	0	3	0.00
IBUFDS	0	0	0	144	0.00
GTPE2_CHANNEL	0	0	0	2	0.00

PHASER_OUT/PHASER_OUT_PHY		0		0		0	
12 0.00							
PHASER_IN/PHASER_IN_PHY		0		0		0	
12 0.00							
IDELAYE2/IDELAYE2_FINEDELAY		0		0		0	
150 0.00							
IBUFDS_GTE2		0		0		0	
2 0.00							
ILOGIC		0		0		0	
150 0.00							
OLOGIC		0		0		0	
150 0.00							
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5. Clocking

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Site Type	Used	Fixed	Prohibited	Available	Util%		
+-----+-----+-----+-----+-----+-----+							
BUFGCTRL	1	0	0	32	3.13		
BUFIO	0	0	0	12	0.00		
MMCME2_ADV	0	0	0	3	0.00		
PLLE2_ADV	0	0	0	3	0.00		
BUFMRCE	0	0	0	6	0.00		
BUFHCE	0	0	0	48	0.00		
BUFR	0	0	0	12	0.00		
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6. Specific Feature

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Site Type	Used	Fixed	Prohibited	Available	Util%		

BSCANE2	0	0	0	4	0.00		
CAPTUREE2	0	0	0	1	0.00		
DNA_PORT	0	0	0	1	0.00		
EFUSE_USR	0	0	0	1	0.00		
FRAME_ECCE2	0	0	0	1	0.00		
ICAPE2	0	0	0	2	0.00		
PCIE_2_1	0	0	0	1	0.00		
STARTUPE2	0	0	0	1	0.00		
XADC	0	0	0	1	0.00		

7. Primitives

Ref Name	Used	Functional Category
LUT6	248	LUT
LUT5	96	LUT
LUT2	83	LUT
FDRE	73	Flop & Latch
OBUF	69	IO
IBUF	67	IO
CARRY4	60	CarryLogic
LUT3	33	LUT
FDSE	1	Flop & Latch
BUFG	1	Clock

8. Black Boxes

Ref Name	Used
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9. Instantiated Netlists

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Ref Name Used

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